

Abdullah Ali

MACHINE LEARNING ENGINEER ·

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Summary

Machine Learning Engineer Intern with experience in designing and deploying intelligent AI agents and RAG systems using advanced LLM orchestration frameworks. Proficient in deep learning (Transformers, CNNs, NLP), data preprocessing, and feature engineering, building robust, production-ready systems with Python, FastAPI, and Docker. Skilled in applying MLOps best practices including Apache Airflow, MLflow, CI/CD pipelines, and AWS (EC2, S3, Lambda, ECS, IAM, CloudWatch) to ensure scalability and reliability. Passionate about creating multi-modal AI SaaS platforms that integrate text, vision, and audio intelligence for real-world impact.

Tools & Technologies

• Languages & Libraries:

Python – SQL – Langchain – LangGraph (Learning) – Tensorflow – Scikit-learn – NumPy – Pandas – Matplotlib – FastAPI – Pydantic – Optuna

• AI & Machine Learning:

Agents – RAG – LLMs – Fine-Tuning – Transformer – NLP – ANNs – Supervised & Unsupervised Learning – Hyperparameter Tuning – Model Evaluation – EDA – Feature Engineering

• MLOps & AWS Cloud:

MLflow – Apache Airflow – Docker – CI/CD – EC2 – S3 – RDS – DynamoDB – Lambda – ECS – EKS – IAM – CloudWatch

projects

YouTube RAG Chatbot | GitHub

- Built an AI-powered Retrieval-Augmented Generation (RAG) system that converts YouTube videos into an interactive chatbot capable of **answering user queries** in real-time using Ollama LLMs.
- Automated transcript extraction, text chunking, embedding, vector search using **LangChain**, FAISS, and Ollama model.
- modular pipeline covering transcript ingestion, chunking, embedding, vector store and contextual answer via Ollama-hosted LLMs.
- Containerized and deployed the RAG-powered FastAPI backend using Docker on **AWS EC2**, enabling scalable, **production ready** inference for YouTube question-answering.

Tools: Python, FastAPI, LangChain, FAISS, Transformers.js, HTML/CSS, YouTube API, Vector Databases, GitHub, **JSON**

Conversational Chatbot | GitHub

- Developed a **GPT2-based** conversational AI chatbot capable of processing large-scale dialogue datasets from Hugging Face.
- Built modular **pipelines** for ingestion, validation, transformation, and training with full schema and artifact tracking
- Integrated **MLflow** for experiment tracking, implemented a modular **FastAPI** backend, Build ui by HTML/CSS for user interaction.
- Containerized and deployed the FastAPI using Docker and integrated with **AWS EC2** for scalable cloud deployment.

Tools: Python, GPT2, Transformers, PyTorch, MLflow, Hugging Face, Pandas, FastAPI, **Docker**, AWS EC2, Git, HTML, CSS

Network Security | Github

- Developed an end-to-end machine learning system to detect phishing websites using 30+ engineered features.
- **Achieved 98% accuracy** with tuned Random Forest among multiple classifiers (Decision Tree, Logistic Regression, Boosting).
- Built **modular pipelines** for ingestion, validation, transformation, and training with full schema and artifact tracking.
- Deployed the solution on AWS EC2 with FastAPI, Streamlit, Docker, GitHub Actions (CI/CD), MLflow, DVC, and S3..

Tools: Python, **FastAPI**, **Scikit-learn**, Streamlit, MLflow, DVC, AWS (S3, EC2), Docker, GitHub Actions, **MongoDB Atlas**, Git, Pandas

Certifications

- **Complete Agentic AI Bootcamp** (*Certificate*)
- **Complete MLOps Bootcamp** (*Certificate*)
- **Machine Learning A-Z** (*Certificate*)

Academics

Virtual University of pakistan

BACHELOR IN COMPUTER SCIENCE AND ENGINEERING

Lahore, Pakistan